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Software Engineer

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📍 Kampala, Uganda

EXPERIENCE

Backend Engineer

Googah Goats Limited

Oct 2025 – Present
Kampala, Uganda

- Architect backend infrastructure using Google AppScript and JavaScript for startup operations.
- Design data workflows and automation systems connecting spreadsheet-based backend to frontend applications.
- Build custom APIs and integration scripts, demonstrating full-stack development capabilities.

Technical Instructor

Neriko Electronics

Oct 2024 – Sep 2025
Kampala, Uganda

- Taught embedded systems programming using C/C++ and Arduino to students.
- Developed curriculum and hands-on labs, demonstrating strong technical communication skills.
- Instructed students in development tools: PlatformIO, TinkerCAD, Fritzing, Proteus, Eclipse, and NetBeans.

PROJECTS

terminalphone – Encrypted P2P Voice over Tor

[View Repository](#)

Bash, Tor Hidden Services, Opus, OpenSSL, FFmpeg

- Single self-contained script providing anonymous, end-to-end encrypted push-to-talk voice and text between parties over Tor hidden services – no servers, accounts, or phone numbers; identities are .onion addresses.
- Adapts Tor for real-time voice via record-then-send streaming, Opus tuned to 16kbps mono for Tor-sized payloads, and OpenSSL encryption with secrets passed via file descriptors; 122+ stars on GitHub.

Amber – Android Reconstruction Pipeline

[View Repository](#)

Kotlin, DEX IR, SSA, JavaPoet, Gradle, Apache 2.0

- Lifts APK and DEX artifacts into typed IR, control-flow graphs, SSA form, and generated source, then synthesises a complete, buildable Android Studio project around the result.
- Implements dex parsing, Java/Kotlin emission, Gradle synthesis, metadata analysis, and evidence-rich diagnostics – turning reverse engineering into a structured, reproducible engineering workflow.
- Supports Compose detection, metadata-aware Kotlin recovery, and full resource pipeline integration; validated end-to-end against real APKs via GitHub Actions CI.

kore – Portable Schrödinger-Poisson Kernel

[View Repository](#)

Rust, PyO3, C ABI, NumPy, RustFFT, Rayon, pykore 0.2.1

- Portable Rust core for Schrödinger-Poisson research workloads; single validated kernel surfaced as Rust crate, `pykore` Python bindings (PyPI 0.2.1), and stable C ABI with generated `include/kore.h`.
- `pykore` NumPy-first API (Grid, Config, State, Kernel) with in-place `complex128` stepping via PyO3, GIL released on compute path; optional HDF5 checkpointing (`checkpoint-hdf5`) for resumable runs, versioned across surfaces.
- Verified by Rust + Python integration tests, strict C11 smoke builds, header-drift checks, and packaging workflows; documented in `docs/index.md` and `paper.pdf`.

Joist – Architectural DSL & Blueprint Compiler

[View Repository](#)

Rust, Domain-Specific Language, Vector Geometry

- Compiler that transforms declarative, AI-writable `.joist` architectural code into professional vector drawing sheets – floor plans, elevations, sections, and multi-view master blueprint sets.
- Pipeline of parser, geometry engine, and validator over shared 3D building geometry (towers, walls, openings, roofs), keeping semantic input and generated drawings consistent across all views.

KaCinema – Netflix-Style Streaming Platform

[View Repository](#)

Spring Boot, Android, Jetpack Compose, REST APIs, Firebase

- Full-stack streaming platform with Spring Boot backend (auth, catalog, watchlist APIs) and native Android client using Jetpack Compose and MVVM with Repository pattern.
- Implements adaptive media playback, caching with Room/DataStore, and REST integration via Retrofit/OkHttp with coroutines/Flow.
- Deployed backend with CI/CD via GitHub Actions; demonstrates scalable cloud + mobile product delivery for content-driven apps.

local-lite – On-Device LLM Inference on Android

[View Repository](#)

Kotlin, ExecuTorch, Qwen 3, Android NDK

- Android app running large language models fully on-device with ExecuTorch and Qwen 3 models – no cloud calls, no data leaving the device.
- Demonstrates mobile AI deployment end-to-end: model packaging, memory-conscious inference, and Kotlin integration over the ExecuTorch runtime.

Flare DAW – Rust Audio Engine & GPUI DAW Workspace

[View Repository](#)

Rust, GPUI, Audio DSP, Real-time Processing, GitHub Actions

- Custom DAW workspace built in Rust with GPUI, featuring a real-time audio engine with lock-free DSP chain (EQ, compressor, reverb) at 128-sample buffers (2.67 ms @48kHz).
- Implements multitrack mixing with pan law, offline WAV export, and project persistence; architected for 60fps UI with <8ms recomposition.
- Validated on Android/iOS toolchains for future mobile DAW port; demonstrates deep audio systems expertise directly relevant to AI-infused workstation work.

youbert – YouTube Live Archiver & VOD Downloader

[View Repository](#)

Go, Real-Time Streaming, Muxing

- Fast, resilient archiver that captures YouTube live streams in real time (or downloads finished VODs) and produces a single muxed `.mkv` file – no browser required.
- Built for reliability under unstable network conditions, with resumable capture and clean CLI ergonomics.

freevie – Free Live TV for Stremio

[View Repository](#)

JavaScript, Stremio Addon SDK, IPTV

- Open-source Stremio addon streaming 500+ live USA, Canada, and Uganda TV channels, auto-updated hourly from the iptv-org catalogue.
- Self-hostable and account-free, with in-addon channel search; deployed and used by real Stremio users.

Calenda – Cross-Platform Calendar App

[View Repository](#)

Kotlin, Jetpack Compose, Room, Swift, SwiftUI, Core Data

- Calendar app built with Jetpack Compose and Room database persistence, then migrated natively to iOS as **Calenda-iOS** using SwiftUI and Core Data.
- Demonstrates platform-native craftsmanship on both sides rather than a lowest-common-denominator port.

HTML-TO-DESIGN – Penpot HTML Conversion Plugin

[View Repository](#)

TypeScript, Svelte 5, Vite, Penpot Plugin API

- Plugin for the open-source design tool Penpot that parses live HTML DOM trees and rebuilds them as editable native Penpot shapes.
- Sandbox-safe plugin architecture with typed UI↔plugin messaging and progressive conversion feedback; 8+ stars on GitHub.

TECHNICAL SKILLS

Languages	Kotlin, Swift, Java, Rust, Go, C/C++, Python, Zig	Async & DI	Coroutines/Flow, RxJava, Combine – Hilt, Dagger, Koin
Android	Jetpack Compose, Navigation, Room, DataStore, WorkManager	Backend & Networking	Spring Boot, Firebase, Supabase – Retrofit, OkHttp, Alamofire, REST
iOS	SwiftUI, UIKit, Core Data	Cross-Platform & On-Device AI	KMP – ExecuTorch, LiteRT (on-device LLM inference)
Architecture	MVVM, Clean Architecture, Repository Pattern	Tooling & CI/CD	Android Studio, Xcode, VS Code – Git/GitHub/GitLab, GitHub Actions, Fastlane

EDUCATION

Associate Degree in Electrical Engineering & Computer Science	2025 – 2027
University of the People	United States
Diploma in Electronics and Electrical Engineering	2025 – 2027
Uganda Institute of ICT	Uganda

LANGUAGES

English	Fluent	Luganda	Fluent	French	Intermediate (B2)
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